

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester V Examination, 2023-24

Paper No. BAS-401: Sampling, Design of Experiment and Quality Control

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Distinguish between sampling and census methods of data collection. [6]
- (b) In the case of simple random sampling without replacement (SRSWOR), prove that the probability of selecting any specified unit in the sample at any draw is n/N , where N is the population size and n is the size of the sample drawn. [8]

Question 2.

- (a) In usual notations, prove that in the case of SRSWOR, the variance of the sample mean is given by:

$$Var(\bar{y}_n) = \frac{N-n}{Nn} S^2 \quad [7]$$

- (b) If $\bar{y}_n$ is the sample mean of a random sample of size n drawn from a population of size N by the method of SRSWOR, prove that it is an unbiased estimator of the population mean, i.e.,

$$E(\bar{y}_n) = \bar{Y}_N \quad [7]$$

Question 3. Define allocation, proportional allocation, and Neyman allocation in stratified random sampling. Derive the expression for the variance of the stratified sample mean estimator:

$$\bar{y}_{st} = \sum_{i=1}^k W_i \bar{y}_i$$

under proportional allocation, where $W_i = N_i/N$ and $\bar{y}_i$ is the sample mean of the i -th stratum drawn using SRSWOR. [14]

Question 4. Define linear systematic sampling. Prove that the systematic sample mean $\bar{y}_{sys}$ is an unbiased estimator of the population mean $\bar{Y}_N$. Also prove that:

$$Var(\bar{y}_{sys}) = \frac{N-1}{N} S^2 - \frac{k(n-1)}{N} S_{wsy}^2$$

where the symbols have their usual meanings.

[14]

Question 5. Describe cluster sampling. In the case of equal cluster sampling, prove that the sample mean $\bar{y}_{cl}$ is an unbiased estimator of the population mean $\bar{Y}_N$. Also, write down the expression of $Var(\bar{y}_{cl})$ in terms of the intra-class correlation coefficient ρ .

[14]

Question 6. Give the layout of a Completely Randomized Design (CRD) and discuss the complete statistical analysis, including the ANOVA table and tests of hypothesis, for a CRD.

[14]

Question 7. Explain the three basic principles of experimental designs: replication, randomisation, and local control. Further, explain the situations in which a Randomized Block Design (RBD) is considered as an improvement over a Completely Randomized Design (CRD).

[14]

Question 8. What do you understand by Control Charts for fraction defective (p -chart)? Explain its construction, setting up of control limits, and discuss the underlying theoretical probability distribution on which these control limits are based.

[14]